HAIYI MEI

Research Fellow | Synthetic Data Engineer | Backend Engineer | Game Developer

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TECH STACK

- Python: 6+ years
- C++: 7+ years
- PyTorch: 6+ years
- Unreal Engine: 4+ years
- Blender: 3+ years
- Git
- Docker
- CI/CD

ACQUAINTED

- Game Engine
- PBR
- Synthetic Data
- SMPLX
- SaaS
- NeRF
- Diffusion
- Differentiable Rendering
- Music (Piano/Guitar/Theory)

EDUCATION

Master's Degree

Shandong University

2019 – 2022

Bachelor's Degree

NJUST

2015 – 2019

- Head of the Symphony Orchestra of School

LANGUAGES

Chineses:	**Native**
English:	**C1**
Japanese:	**A1**

ABOUT ME

I'm a committed algorithm researcher and backend engineer with 3 years of industry experience. Specializing in 3D synthetic data generation for computer vision tasks, I contributed to 9 papers in top-tier conferences. Recently, I've been leading efforts in text/image-to-video generation, contributing to VAE training, dataset preparation, automated annotation, and video acquisition using crawler techniques. I'm familiar with the entire T2V pipeline and passionate about driving innovation in these fields.

EXPERIENCE

Mid-Level Research Fellow | SenseTime

2021 – Present

2024 – Present: Text/Image-to-Video

- Led VAE training, successfully reproducing results comparable to CogVideo's VAE with similar performance.
- Contributed to the data annotation process by supporting the application of textual, visual, and qualitative analyses to ensure high-quality video datasets.
- Developed automated solutions for video content acquisition from various online platforms lacking direct API support.

2021 – 2024: Synthetic Data Generation

- Spearheaded the development of XRFeitoria, a rendering toolbox that streamlines synthetic data generation, contributing to 7 papers in 2 years.
- Mainrole in designing SynBody, a large-scale synthetic dataset with layered parametric human models.
  - Contributed to SMPL-XL, enhancing SMPL-X with hair, garments, textures, etc.
  - Developed a SaaS platform that generated over 10M synthetic annotated images, covering human creation, motion retargeting, scene composition, and rendering.
  - Co-first author of a paper presented at ICCV 2023.
- Proficient in crafting advanced technology demonstration videos.

Research Intern | NLPR, CASIA

2018 – 2021

- Initiated research on synthetic data generation for computer vision tasks.
  - Filed a patent with CNIPA: CN113011334A.
  - Published one paper related to synthetic data.
- Reproduced state-of-the-art methods in video captioning. Code

OPEN-SOURCE PROJECTS

XRFeitoria | Synthetic Data Rendering Toolbox

- Control over engine (UE/Blender) through RPC using system python.
- Render photorealistic images with ground-truth annotations.
- Manage assets/cameras, including import, place, export, and delete.
- CLI tools to render images from a mesh file.
- CI/CD tools for auto-compiling, testing, and distributing Unreal/Blender plugins.
- Streamlined rendering pipeline across various domains, in over 7 projects.

AWARDS

Patents

Hold 4 patents filed with CNIPA

CN108453742B, CN113011334A, CN114429436A, CN115272548A